

KONSTANZ, Germany

WMO: 109290

Lat: 47.6775N

Lon: 9.1900E

Elev: 1458

StdP: 13.94

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.8	18.8	8.5	9.0	15.1	13.5	11.5	21.3	22.7	43.4	19.5	42.2	4.6	0	0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.5	85.4	67.2	82.2	66.1	79.1	64.7	69.1	81.2	67.5	78.5	66.2	76.5	4.4	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.0	97.6	73.5	63.6	93.0	72.4	62.3	88.8	71.1	34.0	81.6	32.7	78.6	31.6	76.6	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.7	13.9	11.7	DB	12.3	90.6	7.6	2.6	6.8	92.5	2.4	94.1	-1.9	95.5	-7.4	97.4	(4)
			WB	11.5	72.1	7.3	1.3	6.2	73.1	1.9	73.9	-2.2	74.6	-7.5	75.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.7	32.6	33.6	41.7	47.9	56.6	62.7	67.7	66.5	60.1	51.0	40.1	35.3
	DBStd	14.01	8.23	7.47	6.55	6.12	6.84	6.73	5.58	5.57	5.53	5.92	6.31	6.49
	HDD50	2224	541	461	268	112	24	1	0	0	1	60	300	456
	HDD65	5866	1006	881	724	514	266	121	38	51	164	435	746	920
	CDD50	2126	0	0	9	47	229	383	548	510	304	91	4	1
	CDD65	293	0	0	0	0	5	54	120	97	17	0	0	0
	CDH74	2189	0	0	0	3	75	423	943	654	90	1	0	0
	CDH80	527	0	0	0	0	3	99	260	159	6	0	0	0
Wind	WSAvg	4.1	4.7	4.7	5.0	4.7	4.1	4.1	3.6	3.4	3.5	3.7	3.9	4.3
Precipitation	PrecAvg	32.9	1.8	1.8	2.1	2.4	3.5	4.0	3.9	3.6	2.8	2.4	2.3	2.6
	PrecMax	41.2	5.3	4.4	5.7	5.4	7.3	6.9	8.0	7.6	6.3	4.5	5.7	6.3
	PrecMin	24.3	0.1	0.6	0.6	0.3	0.6	1.2	1.0	0.6	1.3	0.4	0.0	0.1
	PrecStd	4.3	1.3	0.9	1.2	1.4	1.7	1.4	1.7	1.9	1.3	1.2	1.3	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.3	56.6	68.5	72.5	79.7	87.0	89.9	89.0	80.1	71.4	59.2	53.6
		MCWB	46.4	47.6	53.8	54.8	60.8	66.7	68.7	68.1	65.1	60.5	51.1	47.9
	2%	DB	49.0	50.4	59.5	67.4	76.0	82.7	86.2	84.4	76.6	65.5	53.9	50.6
		MCWB	44.3	44.3	48.4	52.4	59.3	65.9	67.6	66.9	64.5	57.9	49.9	46.7
	5%	DB	45.8	46.4	55.1	63.5	73.0	78.8	82.7	80.8	73.2	61.8	51.0	47.7
		MCWB	42.2	42.3	46.1	50.7	57.9	64.2	66.5	65.9	62.8	55.7	47.8	44.4
	10%	DB	42.8	43.4	51.3	59.4	69.2	75.1	79.4	77.4	69.9	58.7	48.4	44.1
		MCWB	40.2	40.2	44.6	48.9	56.4	62.4	65.4	64.6	61.0	54.2	45.7	41.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.3	48.2	53.3	56.1	63.1	70.0	71.3	70.6	67.1	61.4	52.8	50.1
		MCDB	51.8	53.4	66.9	69.7	75.4	81.0	85.6	84.0	76.9	69.0	56.5	53.2
	2%	WB	44.7	45.2	49.6	53.5	60.6	67.6	69.4	68.4	65.2	59.0	50.4	46.8
		MCDB	48.3	49.1	57.6	65.3	72.7	77.7	82.3	80.2	74.4	63.5	53.5	49.6
	5%	WB	42.4	42.8	47.3	51.7	59.3	65.7	67.9	67.0	63.6	56.7	48.2	44.3
		MCDB	45.3	46.3	53.3	60.7	69.8	75.4	79.3	77.6	71.3	60.3	51.0	47.3
	10%	WB	40.3	40.4	45.4	50.0	57.6	63.6	66.4	65.7	61.9	54.7	46.1	41.8
		MCDB	42.7	43.1	50.3	58.2	66.7	73.2	77.3	75.2	68.1	58.0	48.2	44.1
Mean Daily Temperature Range	5% DB	MDBR	8.0	10.3	14.1	16.5	17.8	17.3	18.5	17.9	15.7	11.7	8.4	7.2
		MCDBR	11.8	15.0	20.7	25.0	24.7	23.2	24.2	23.4	20.1	15.9	12.0	11.3
		MCWBR	8.0	9.8	11.2	12.5	11.3	9.3	10.0	9.3	9.7	8.7	8.5	8.4
	5% WB	MCDBR	11.4	14.1	18.1	21.7	21.3	21.7	21.6	21.1	18.9	15.1	11.4	10.8
		MCWBR	8.3	9.8	10.4	11.2	10.2	8.9	9.6	9.1	9.4	8.7	8.8	8.5
Clear-Sky Solar Irradiance	taub	0.293	0.317	0.352	0.389	0.401	0.407	0.409	0.402	0.376	0.347	0.316	0.288	
	taud	2.443	2.377	2.349	2.278	2.284	2.305	2.293	2.330	2.382	2.451	2.486	2.451	
	Ebn at Noon	252	267	274	273	274	272	269	266	263	253	240	239	
	Edh at Noon	21	28	33	39	41	40	40	37	32	25	20	18	
All-Sky Solar Radiation	RadAvg	351	611	994	1367	1586	1817	1744	1524	1138	710	390	287	
	RadStd	39	86	134	195	161	165	158	156	118	76	47	45	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (111 neighbors)		N/A	N/A	N/A	N/A	+0.77	+0.67	N/A	N/A	+97	+25

Nomenclature: See separate page